4. THE PROPERTIES OF THE HIGHWAY TRAFFIC

The mobility created by vehicles and pedestrians, under some rules, generates highway traffic. The traffic flow term is used to show various displacements on highways. When designing a highway, before and after the construction, knowing the properties of the traffic, changes that occur, and the relations between the variables is important for choosing a safe and economical solution.

4.1 The change of the traffic

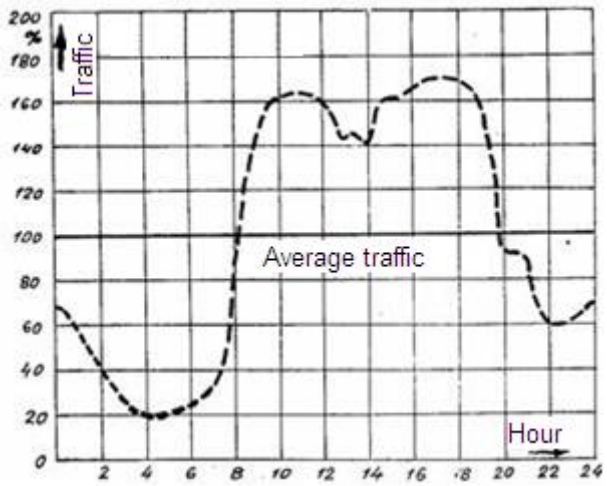
Highway traffic changes according to geophysical, social, economic, and dynamic factors. These changes are usually periodical, or in other words, they are repeated on specific periods. For example, the amount of traffic starts to increase with the sunrise, starts to decrease with the sunset, and almost perishes with the night. For this reason, the situation of the sun has a great impact on the periodical changes.

The volume of traffic decreases in winter, and increases in summer when the cold continental climate is effective. In this situation, the temperature is effective on the change. Weekends and annual vacations have different effects on different places. In agricultural zones, during the harvest season, traffic increases. In industrial and commercial zones and touristic zones, changes can be seen in various periods.

For this reason, the properties of the change in the amount of highway traffic during different periods have to be researched separately.

4.1.1 Hourly change of the traffic

The hourly change represents the changes in the amount of traffic during the hours of the day. For example, on an urban highway, the volume of the traffic starts to increase around 07:00 and reaches the maximum value around 08:00-09:00. In the evening, the increase starts around 17:00 and reaches the maximum value around 18:00-19:00. After that it starts to decrease, contrary to the morning traffic. This maximum traffic in the morning or evening hours is called the peak hour traffic.

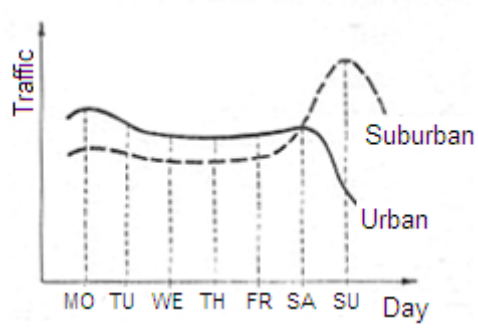


On rural roads, a similar situation occurs, but it is both lower than on urban roads, and the peak hours shift a bit earlier. On urban highways, on normal work days, morning traffic is towards the city center, and home-based work and home-based school trips are predominant. During the daytime, work-based work and home-based shopping type trips are predominant. At evening hours, work-based home trips are predominant, while work-based entertainment, work-based shopping, and home-based entertainment trips are also seen. The direction of the traffic is usually from the city center to the environment.

Work-home trips are a little bit less than the morning because some people attend to other activities before returning to their houses. A design must be made to take into account the trip properties listed above, especially urban highways. During recent years, the peak hour has a tendency to shift a little bit earlier at some metropolis due to the increasing population and car ownership. Due to the increasing traffic demand, people leave their houses earlier. In the evening, people can shift their leave hours a bit earlier or later, according to this change.

4.1.2 Daily change of the traffic

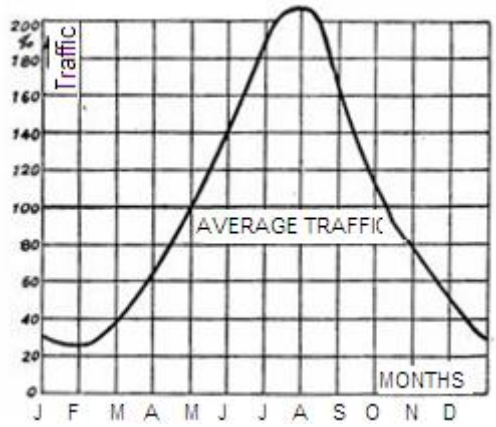
The daily change represents the change in the amount of traffic between the days of the week. On urban highways, as the Monday is the first day of the week, the traffic is denser than the other days. From Tuesday to Friday, the traffic is almost at the same level. Because people usually rest on weekends, urban highway traffic decreases. On the contrary, traffic on suburban highways increases because of vacation trips. As seen in the figure below, traffic on suburban highways is less than urban traffic on weekdays. Friday afternoon, the suburban traffic starts to increase. Weekend traffic shows an inversely proportional change on the suburban and urban highways.



4.1.3 Monthly change of the traffic

The monthly change represents the change in the amount of traffic in the months of a year. As the hourly traffic follows the sun, the monthly traffic is also affected by the position of the sun. Traffic increases in summer in warm climates. The traffic values of May and October represent the average in those places. Traffic decreases in summer in hot climates.

On urban highways, traffic decreases in summer because home-based-school trips are at a minimum level. Also, as people spend their holiday in the countryside, traffic also decreases in summer on urban highways. The figure below shows the seasonal change in the traffic on suburban highways.



4.1.4 Annual change of the traffic

The amount of traffic shows some changes from year to year. This situation, which is usually the result of the population and employment increase, has an increasing trend. Construction of traffic attractions or production centers like commercial centers, industrial installations, ports, and airports may cause an annual change.

4.1.5 Aperiodic change of the traffic

Traffic may change due to extraordinary events besides normal changes. Disasters like earthquakes, floods, hurricanes, or wars may be listed as the causes of the aperiodic change in traffic. These special cases need to be taken precautions without taking certain rules.

4.2 Main relationship of the traffic

Some models have been developed to explain the existing relations between the variables of highway traffic. A model that shows the relationship between the volume, density, and speed variables is given.

Main variables of the traffic: These variables are acquired by various methods, but they are not used independently. The other variables can be acquired by using the main variables length **(X)** (m, km), time **(T)** (sec, min, h), and traffic unit **(N)** (veh).

Derived variables of the traffic: They are the variables derived from the main variables of the traffic to explain the traffic flow. They can be listed as the volume **(q)** (veh/sec, veh/h), density **(k)** (veh/km), and speed **(u)** (m/sec, km/h).

Flow rate: The number of vehicles that pass from a highway section in a period of time is the flow rate. The period of time can be taken as the intervals like 5, 10, or 15 minutes.

Traffic volume: The number of vehicles that pass from a highway section within an hour is the traffic volume. It can be calculated for the whole highway section or a single lane. It is shown as q=N/T.

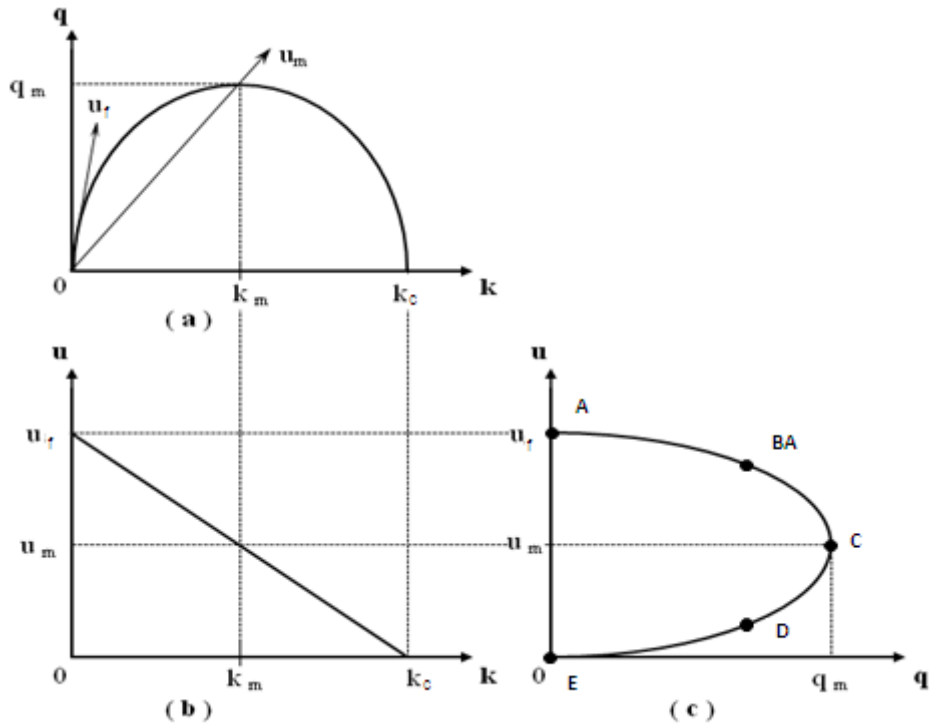
Traffic density: The number of vehicles found on a specific length of the road at an instant is the traffic density. It can be calculated for the whole highway section or a single lane just like the traffic volume. It is shown as k=N/X.

Speed of the traffic flow: The distance covered by the traffic flow that is constituted by the vehicles on the unit time is the speed of the traffic flow. It is shown as u=X/T.

According to the definitions above, the main relationship of the traffic can be written as:

$$q = k \cdot u$$

The relation between the three variables in the main relationship can be explained using the figures below.



Although there are various models in the literature, the one known as the Greenshield Model is useful in explaining the relationship between the variables. Some assumptions and idealizations have been made for drawing the curves. The main curves of the traffic flow can be explained as:

As the traffic density increases on a highway, the traffic flow also increases. This increase continues until the capacity of the highway (the maximum number of vehicles that can pass, $\mathbf{q_m}$) is reached. After this point, as the demand exceeds the capacity, density increases while the traffic flow decreases, and this decrease continues until the $\mathbf{k_c}$ of the congestion density. After the demand for the related highway section exceeds the capacity, the mobility of the highway or, in other words, the number of vehicles that can pass decreases. The slope of the line that connects any point to the origin gives the speed of the traffic flow at that point. The slope of the tangent of the origin is the free speed ($\mathbf{u_f}$). The slope of the line that connects the point (0,0) and point ($\mathbf{q_m}$, $\mathbf{k_m}$) is the speed of the flow if the capacity is reached. After the density in the capacity ($\mathbf{k_m}$) congestions are seen in the traffic flow. At congestion density

($\mathbf{k_c}$), although there are vehicles on the highway, the flow does not move (or moves very slowly) $\mathbf{q \rightarrow 0}$.

As the number of vehicles increases on a highway, the effect of vehicles on each other also increases, and as a result, the speed decreases. In the beginning, there are very few vehicles on the road; in other words, the effect of vehicles on each other is low; drivers are free to select the speed they want (they can drive at high speeds if they want). The speed in this condition is named the free speed ($\mathbf{u_f}$). The speed obtained from the density of the capacity $\mathbf{k_m}$ is $\mathbf{u_m}$. On $\mathbf{k_c}$, $\mathbf{q \rightarrow 0}$, so $\mathbf{u \rightarrow 0}$.

The **AB** region of the curve shows the free flow on which the free speed can be preserved. As the traffic flow increases, the speed decreases. The decrease in the velocity continues until the capacity value ($\mathbf{q_m}$). On velocities just over and under the capacity value (**BD** region), traffic flow is unstable. As the traffic exceeds the capacity, forced flow starts to occur. In this situation, there are more stop-and-goes, and queues are seen (**DE** region). As we get closer to **E,** the traffic starts to congest, and the flow almost stops.

Peak hour factor

The traffic volume is the number of vehicles that can pass from a highway section in an hour, as explained before. However, as the flow is continuous in this 1 hour, it is not uniform. Usually, it decreases or increases within different time intervals of this 1 hour, or in other words, it oscillates. The flow rate term is used to show this oscillation. The time interval for the flow rate is usually 15 minutes (the calculated value is converted into the hourly value by calculation). The ratio between the measured traffic flow and the calculated traffic flow is named the **peak hour factor (PHF)**. It can be formulated as:

$$PHF = \frac{q}{q_m} \qquad\qquad q_m = \frac{N_T \quad 60}{T}$$

q_m : Maximum flow rate (veh/h).

N_T : The number of vehicles that pass from a highway section in the T period of time.

T : The time period used on the flow rate (min).

As an example, 1500 vehicles are counted in an hour on a highway section, while the maximum value counted during the 15-minute intervals of this hour is 400 vehicles. The flow rate is calculated as:

$$q_m = \frac{N_T \quad 60}{T} = \frac{400 \quad 60}{15} = 1600 \ \text{veh/h}$$

As it can be seen, the measured and the calculated flow rate values can vary. Accordingly, if the peak hour factor is calculated,

$$PHF = \frac{1500}{1600} \quad 0{,}94$$

is obtained.

The most desired value of the peak hour factor is 1. The uniformity disrupts, and oscillations on the traffic increase when diverged from this value.

4.3 Project traffic

The traffic volume that is used to design a highway is named as the **project traffic**. In highway design, the usage of accurate and realistic project traffic is very important. According to the nature of the highway, it provides transportation to a zone as well as generates economic, social, and cultural improvement around its route. The accurate projection of the traffic that emerged as a result of this improvement is important in making an investment.

Properties about the annual, monthly, daily, and hourly change of the traffic have to be known to reflect the accuracy of the project traffic. Estimation is done for a specific period of time. The time period in which the highway is expected to provide utility is named project life or **service life**.

The project life that is especially considered in the design of suburban highways is between 10-25 years. Generally, it is chosen as 20 years. If the project life is chosen below 10 years, the highway may lose its serviceability due to congestion. If it is chosen over 25 years, the socio-economical changes and the changes in the parameters like population, employment, and land use may not be estimated well enough.

4.3.1 Annual average daily traffic (AADT)

Average daily traffic is the average number of vehicles that pass from a highway section during the day. A time period, like a month or year, is chosen for the traffic study, and the total traffic during this time period is divided into the number of days to calculate the average daily traffic. If this time period is chosen as a year, the total traffic is divided into 365, and the value obtained is named as the **annual average daily traffic (AADT)**.

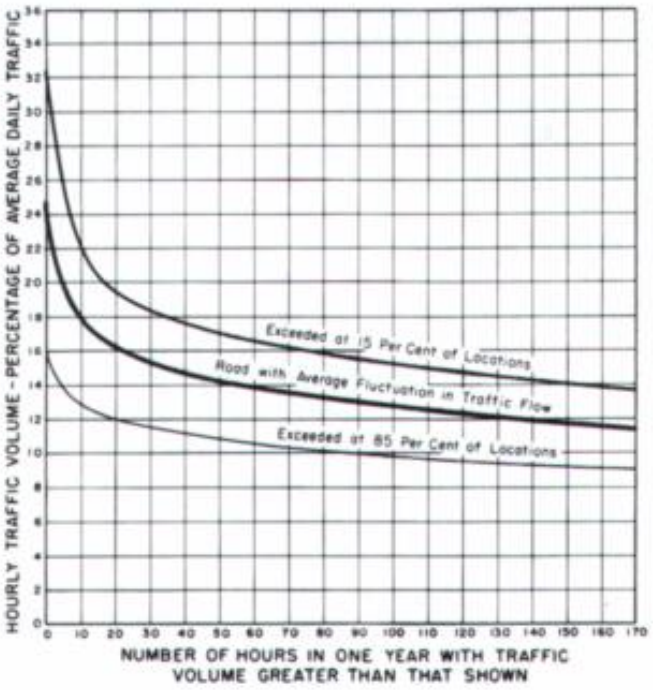
The AADT value calculated gives an idea about the traffic on the highway, but it does not reflect the changes in smaller periods. It is not used as a project traffic. However, it is appropriate to use for the side roads, which carry a low amount of traffic and do not differ much. So, the hourly traffic on which changes during a day must be calculated.

4.3.2 Design hour traffic (30th hour traffic)

How to determine the design hour traffic, which will have an impact on selecting the highway design standards, is an important question. If the design standards are chosen for the experienced highest hourly traffic, the highway would be designed more than needed, and it becomes inactive for almost the whole time. As it is over-invested, it will not be an economical solution.

If the design standards are chosen for the average daily traffic, on many hours during a year, traffic will be higher than the average, and congestion will be seen.

To overcome this problem, a relation between the calculated AADT and the observed daily traffic has been determined. To show this relation, the AADT, and the hourly traffic (q) values are found for a highway, and the traffic values found are calculated using the percentage of the AADT. Then, the values obtained are listed as high or low. On the horizontal axis, the hourly traffic values, and on the vertical axis, the percentage of these values on AADT are written, and the figure shown is drawn.



On the curves, 30th-hour traffic is seen to be a critical value; on the left side, the curve steepens, while on the right side, it leans. This means that there are many hourly traffic lower than 30th-hour traffic while there are few more than that. In other words, during a year, the traffic will be higher than 30th-hour traffic 29 times and more times lower than that. In a year, the expected congestion will be less while the free-flowing traffic will be experienced more. Studies show that the 30th-hour traffic is around 12~18% of the AADT on suburban highways while 8~12% of the AADT on urban highways.

If an average calculation factor K is to be determined as a percentage of the ADDT, on suburban highways, K=1/6, and on urban highways, K=1/8 can be used. K factor is low in residential zones because the traffic is relatively balanced except at night times. On suburban highways, rural zones, or places where AADT is low, the K factor increases.

This traffic amount calculated according to a percentage of the AADT is accepted as the applicable maximum amount and named the **design hour traffic (DHT)**:

$$DHT = (AADT) \cdot K$$

4.3.3 Traffic forecast

The calculation of the traffic amount expected for a time interval is named the **traffic forecast**. The values obtained on traffic estimation are AADT and DHT.

Traffic forecasting is a hard subject to understand, evaluate, and conclude because it is related to effects such as changes in the population, changes in the socioeconomic structure, improvements in transportation technologies, and changes in people's behavior regarding transportation systems. Many mathematical models and methods have been evaluated for traffic forecasts. Realistic forecasts can be made using these methods, especially if they are based on true and consistent data.

The traffic amount expected to pass from a highway at the end of its service life has several components.

The expected traffic can be expressed as:

$$T_n = T_0 + T_r$$

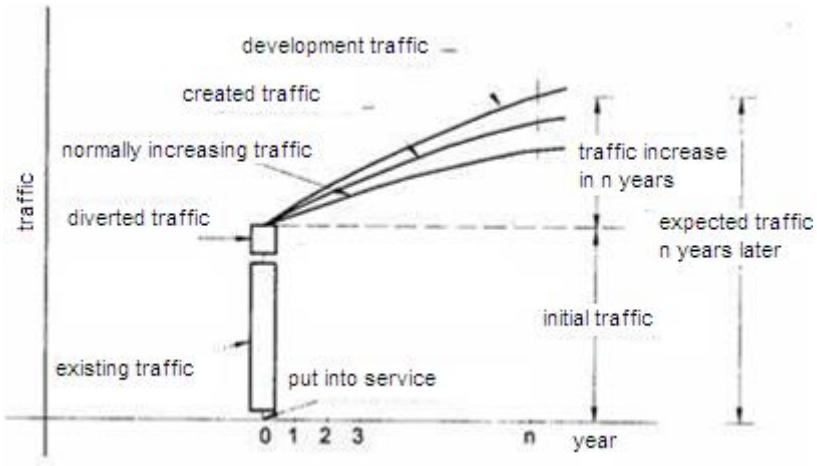
T_n : Expected traffic (At the end of the service life, *n* years)

T_0 : Initial traffic

T_r : Increasing traffic (with an *r* increment ratio)

The components of the initial traffic are the existing traffic and the diverted traffic. The components of the increasing traffic are the normally increasing traffic and the development traffic. The last two components are sometimes accepted to exist within the first two.

The components of the expected traffic can be shown in the figure below.



Initial traffic: The traffic that is created by the people who would use a newly constructed highway or the users of a present but about to-be-improved highway is named the initial traffic.

Present traffic: The traffic found by traffic studies and countings on an existing highway between the two centers is the present traffic. If there is not an existing highway between the two centers, origin-destination (O-D) studies are made to estimate the traffic.

Diverted traffic: The traffic diverted from other highways to the highway to be constructed is the diverted traffic. The possibilities that the new highway provides are the main reason for the diverted traffic. Hence, people are very sensitive to distance or time gain, decrease vehicle costs, and increase safety and comfort increase.

Normally increasing traffic: The traffic increases with the population of the zone, the number of vehicles, and mobility (average number of trips done each day by vehicles), which is the normally increasing traffic. The determination of the normally increasing traffic is very hard because the parameters it depends on are many and different.

However, the expected traffic after *n* years can be calculated using the compound interest equation by using a similar increase in the past years:

$$T_n = T_0 \cdot (1+r)^n$$

r : Annual average traffic increment ratio

n : Service life

The normally increasing traffic amount can be written as:

$$T_r = T_n - T_0$$

Created traffic: The people have repressed their transportation needs due to several negative effects. As the conditions improve, people start to use the transportation system in order to satisfy their needs. The traffic created by the construction of a new highway or the improvement of an existing highway's triggering these needs is the created traffic. Created traffic emerges within a few years after the highway is put into service.

Development traffic: It is already stated that a highway boosts the route that it passes from. The traffic generated by the land use changes, increment of some activities after a highway is put into the service, is named the development traffic.

4.4 Highway capacity

The capacity term is used to state the maximum level that can be benefitted from an object with specific properties, under specific conditions, purposive to a specific subject. Accordingly, examples can be given, such as the capacity of a container, a class, or a factory. The capacity of a highway can be stated as the maximum number of vehicles that can pass in a single or double direction at a specific time period from a section with specific properties under specific conditions.

A highway is being designed to give correct and healthy decisions on subjects like under which conditions and how a highway is going to serve during its life, how many vehicles can pass, what must be the width of it or how many years the service life should be, the capacity has to be known and evaluated.

There are many resources for capacity analysis. The most referenced resource is the ***"Highway Capacity Manual"*** (Special Report 209, Transportation Research Board). The first version of this reference was published in 1950. 2000 versions of it have been used for a long time. The last version was published in 2010.

Theoretical capacity: The maximum number of vehicles that can pass from a highway section within an hour in a single or double direction under the ideal road, traffic, and control conditions is named the theoretical capacity.

The theoretical capacity is usually not reached, but it shows the maximum level that the highway can reach. The ideal conditions on the definition for two lanes and two direction highway can be listed as:

Project speed on two-lane highways is 100 km/h, and on multi-lane highways, 110 km/h or more.

The lane width is 3,65 m or more.

The banquet width is 1,80 m or more.

There are no sections on which the overtake is restricted (overtake distance is more than 450 m).

The turning, joining, or leaving vehicles have no effect on the onward traffic.

The only vehicle in the traffic composition is the automobile.

The distribution of the traffic to the directions is equal (%50/%50).

The capacity value on the ideal conditions listed above for a two-lane, two-direction highway is 2.800 automobiles/hour in both directions. On multi-lane highways, for the project speed of 100 km/h, 2.000 automobiles/hour; for the project speed of 80 km/h, 1.900 automobiles/hour on each lane.

Practical capacity: The capacity value on dominant road, traffic, and control conditions, without big congestions or accidents on the traffic flow; the drivers have no restraints on

speed selection, vehicle following distance determination, or no restrictions on vehicle overtaking possibilities is the practical capacity. It is hard to determine dominant roads, traffic, and control conditions based on practical capacity. By taking the changing conditions into account as an indicator of the practical capacity, the level of service concept is developed. Practical capacity is lower than the theoretical capacity.

There are many factors affecting the capacity of highways. They come along with the road, traffic, and control conditions in the capacity definition. Also, the behaviors of the drivers and the pedestrians on the highway affect the capacity.

***Road conditions*:** Project speed, the situation of the land and environmental properties of the highway route, number of lanes, lane and banquet width, properties about the horizontal and vertical geometry (grade, curve radii, etc.), sight conditions and distances, the pavement condition, and properties can be listed as the road conditions.

Traffic conditions: The vehicle composition (vehicle types and distributions) and the distribution of traffic on lanes and directions can be listed as the traffic conditions.

Control conditions: The distributions and density of the at-grade junctions on the highway, types and control styles of the intersections, properties of the horizontal and the vertical signings, parking possibilities, the type and efficiency of the controls made by the police, current laws and regulations can be listed as the control conditions.

Driver and pedestrian behavior: The level of obeyance of the drivers and pedestrians to the traffic rules, the driver's knowledge of the highway that they use, their experience, lane use properties of the drivers (erroneous lane use, often lane change habits, etc.), passenger drop-off and pick-up characteristics can be listed as the driver and the pedestrian behavior.

4.5 Level of service

As stated in the definition of practical capacity, it is hard for drivers and pedestrians to exactly determine the changing road, traffic, and control conditions. The level of service concept is developed to better understand the qualitative conditions of the users (drivers and pedestrians). The level of service, in general, is the condition of the given lane or platform under various traffic volumes.

The factors affecting the level of service, which can be classified as A, B, C, D, E, and F, are speed, travel time, maneuver freedom (lane change and overtake), vehicle following distance, the interruptions on the traffic flow because of the intersections and signaling, safety and user comfort. As the qualitative parameters are hard to measure, to determine the level of service, the speed and service volume/theoretical capacity ratio (volume/capacity rate in short) is used. Level of service A represents the best situation of the highway with respect to the operating conditions, while level of service F represents the worst situation. In other words, the level of service A corresponds to the free flow, while the level of service F corresponds to the forced flow.

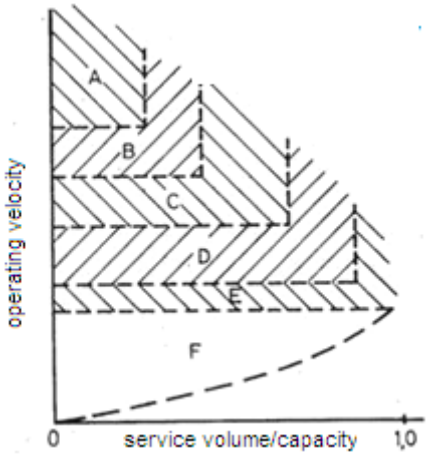
Operating speed: The maximum speed that can be selected by the drivers on a highway section under proper weather and dominant traffic conditions without exceeding the project

speed is the operating speed. In other words, the maximum commercial speed under given dominant conditions.

Service volume (Service flow rate): The maximum number of vehicles that can pass from a lane or the whole highway under the conditions that the selected level of service provides.

Another concept that has to be taken into account is the **delay percentage**. The delay on highways is the loss of time that the vehicles moving in the flow are exposed to because of being unable to select the velocities or to make the manouevres they want. As the traffic density increases, the speed decreases; thus, the delay percentage increases. As it is an indicator of the worsening operating conditions of a highway, its increase shows that the level of service will decrease.

As the level of service and the capacity properties are similar on urban and suburban highways, two and multi-lane highways, motorways, and intersections, there are some differences in the calculation of the values.

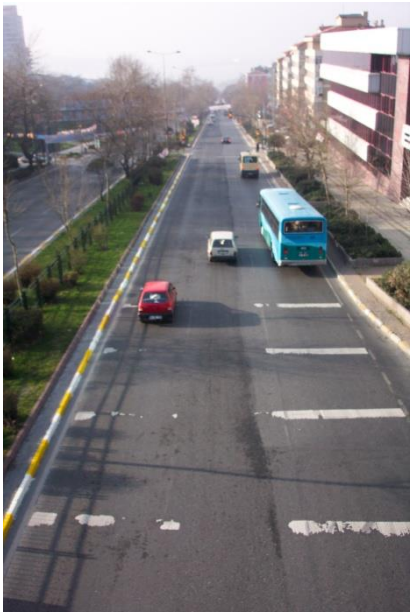


Levels of service shown on the figure can be briefly explained as:

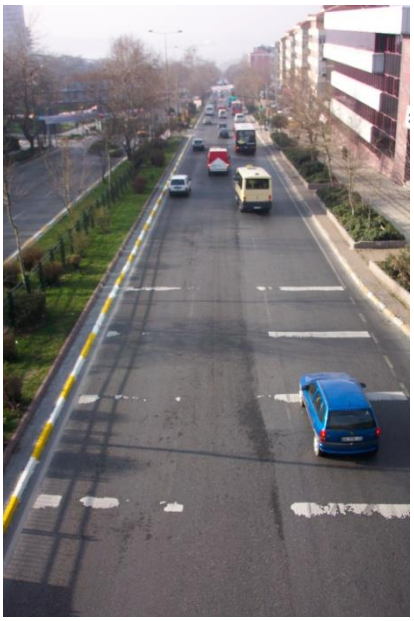
Level of service A: Shows the free flow conditions on which the drivers are not affected by each other during the drive. Drivers are free to select their velocities and make overtake and lane change maneuvers unless there's a speed or sight distance constraint. Travel is very comfortable. The average speed is around 100 km/h. The speed at the level of service A is named the **free speed**. The volume/capacity ratio is lower than 15%. Service volume is below 420 veh/h as the sum of the two directions. The delay percentage is below 30%.



Level of service B: Traffic flow is steady on this level. There is not an important constraint on the speed selection and manouevre freedom of the drivers. An important speed decrease is not monitored. Travel comfort is good, although not as good as the free flow. The average speed is around 90 km/h, the average delay is below 45%, the volume/capacity ratio is below 25%, and the service volume for the two directions is below 750 veh/h.



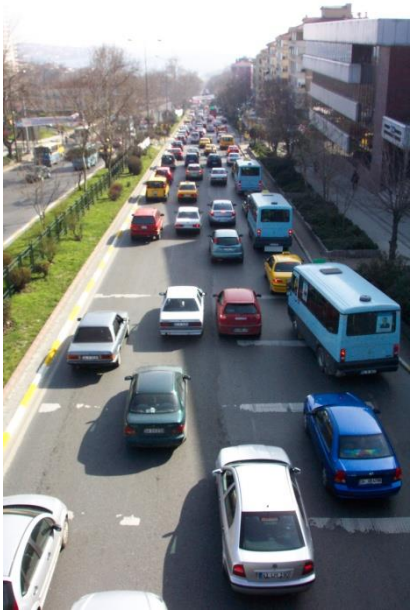
Level of service C: Although the steady condition of the traffic flow continues, the effect of the low-speed vehicles starts to be felt. So, the speed selection and maneuver freedom of the drivers are constrained. Travel comfort and speed start to decrease. This condition shows that traffic congestion may occur. The average speed is around 85 km/h, the maximum value of the delay is 60%, the volume/capacity ratio is around 40% or less, and the service volume for two directions is around 1.200 veh/h or less.



Level of service D: The traffic flow is closer to the unsteady state. The speed selection and the maneuver freedom of the drivers are constrained as an effect of other drivers. Travel comfort and velocities continue to decrease. The effect of overtaking, merging, and diverging vehicles is high. Also, congestion and queuing started to appear. The limit of the average speed is 80 km/h, and the delay is around 75%. The volume/capacity ratio is around 65%, and the service volume for two directions is around 1.800 veh/h or less. This level of service is acceptable for suburban highways.



Level of service E: Traffic flow is unsteady. The volume/capacity ratio is very close to 100%. The drivers' ability to select their speed or to maneuver is totally constrained. Traffic flow is almost congested, and the maneuver of the vehicles feeds this congestion. The average speed is below 80 km/h and on long ascending sections below 40 km/h. Delay is over 75%. The service volume for two directions is around 2.800 veh/h and very close to the capacity of the road.



Level of service F: Volume/capacity ratio is over 100%. In other words, the demand for the road is higher than its capacity. The delay is very close to 100%. As the traffic is totally congested, it tries to flow forcefully. Level of service F shows the limit case on which the road can no longer serve.



4.5.2 The factors affecting the level of service

Some adjustments have to be made to calculate the service volume on various service levels to reflect dominant road and traffic conditions.

The level of service on two lane two direction highways:

$$Q_{SLi} = 2.800 \cdot (Q/C)_i \cdot f_d \cdot f_w \cdot f_{HV}$$

Q_{SLi} : The service volume as the sum of the two directions for level of service (i) under dominant road and traffic conditions (veh/h)

$(Q/C)_i$: The volume/capacity ratio for the level of service (i).

f_d : The direction distribution factor of the traffic.

f_w : The adjustment factor about the lane and banquet width.

f_{HV} : The adjustment factor for heavy vehicle traffic. For its calculation, the following formula is used:

$$f_{HV} = \frac{1}{\left[1 + P_T(E_T - 1) + P_B(E_B - 1)\right]}$$

P_T : The percentage of trucks on the traffic.

P_B : The percentage of busses on the traffic.

E_T : Unit vehicle equivalent of the truck.

E_B : Unit vehicle equivalent of the bus.

The level of service criteria on two lane two direction suburban highways

Level of Service	Dela Percentage (%)	Average Velocity (km/h	Volume/Capacity Ratio					
			The short section percentage on which the sight distance is below 450m					
Flat			0	20	40	60	80	100
A	< 30	> 93	0,15	0,12	0,09	0,07	0,05	0,04
B	< 45	> 88	0,27	0,24	0,21	0,19	0,17	0,16
C	< 60	> 83	0,43	0,39	0,36	0,34	0,33	0,32
D	< 75	> 80	0,64	0,62	0,60	0,59	0,58	0,57
E	> 75	> 72	1,00	1,00	1,00	1,00	1,00	1,00
F	100	< 72	-	-	-	-	-	-
Undulating								
A	< 30	> 91	0,15	0,10	0,07	0,05	0,04	0,03
B	< 45	> 86	0,26	0,23	0,19	0,17	0,15	0,13
C	< 60	> 82	0,42	0,39	0,35	0,32	0,30	0,28
D	< 75	> 78	0,62	0,57	0,52	0,48	0,46	0,43
E	> 75	> 64	0,97	0,94	0,92	0,91	0,90	0,90
F	100	< 64	-	-	-	-	-	-
Mountainous								
A	< 30	> 90	0,14	0,09	0,07	0,04	0,02	0,01
B	< 45	> 86	0,25	0,20	0,16	0,13	0,12	0,10
C	< 60	> 78	0,39	0,33	0,28	0,23	0,20	0,16
D	< 75	> 72	0,58	0,50	0,45	0,40	0,37	0,33
E	> 75	> 56	0,91	0,87	0,84	0,82	0,80	0,78
F	100	< 56	-	-	-	-	-	-

The service volume values on two lane two direction suburban highways

The service volume values on two lane two direction suburban highways

Level of Service	Dela Percentage (%)	Average Velocity (km/h	Unadjusted service volume (both directions) (veh/h)					
			The short section percentage on which the sight distance is below 450m					
Flat			0	20	40	60	80	100
A	< 30	> 93	420	336	252	196	140	112
B	< 45	> 88	756	672	588	532	476	448
C	< 60	> 83	1.204	1.092	1.008	952	924	896
D	< 75	> 80	1.792	1.736	1.680	1.652	1.624	1.596
E	> 75	> 72	2.800	2.800	2.800	2.800	2.800	2.800
F	100	< 72	-	-	-	-	-	-
Undulating								
A	< 30	> 91	420	280	196	140	112	84
B	< 45	> 86	728	644	532	476	420	364
C	< 60	> 82	1.176	1.092	980	896	840	784
D	< 75	> 78	1.736	1.596	1.456	1.344	1.288	1.204
E	> 75	> 64	2.716	2.632	2.576	2.548	2.520	2.520
F	100	< 64	-	-	-	-	-	-
Mountainous								
A	< 30	> 90	392	252	196	112	56	28
B	< 45	> 86	700	560	448	364	336	280
C	< 60	> 78	1.092	924	784	644	560	448
D	< 75	> 72	1.624	1.400	1.260	1.120	1.036	924
E	> 75	> 56	2.548	2.436	2.352	2.296	2.240	2.184
F	100	< 56	-	-	-	-	-	-

Direction distribution factors of the traffic (f_d)

$V_p \geq 100$ km/h; f_d=1,0; f_w=1,0; f_{HV}=1,0 are taken.

Distribution on directions	100/0	90/10	80/20	70/30	60/40	50/50
Adjustment factor	0,71	0,75	0,83	0,89	0,94	1,00

Availible banquet width (m)*	b=3,7 m.		b=3,3 m.		b=3,0 m.		b=2,7 m.	
	Level of Service		Level of Service		Level of Service		Level of Service	
	A-D	E	A-D	E	A-D	E	A-D	E
>=2,0	1,00	1,00	0,93	0,94	0,84	0,87	0,70	0,76
1,0	0,89	0,96	0,82	0,91	0,75	0,84	0,63	0,73
0	0,70	0,88	0,65	0,92	0,58	0,75	0,49	0,66

*If banquet widths are various, their average is taken.

Another subject that has to be taken into account in the service volume and capacity analysis is the **unit vehicle equivalents** of the heavy vehicles in the traffic composition. Heavy vehicles occupy more space than automobiles on highways, and their acceleration, deceleration, and ability to preserve their velocities on ascending sections and maneuvers are weaker than automobiles. This characteristic negatively directly affects the capacity of ascending sections.

The unit vehicle equivalent (uv) used to take this effect of heavy vehicles into account shows the equivalence of heavy vehicles on the number of automobiles in terms of size, speed, and ability to maneuver. According to the situation of the land and the change in the level of service, on two-lane, two-direction highways, trucks have an effect equal to 2~12 uv. Busses have an effect of 1,6~6,5 uv on the same condition.

Unit vehicle equivalents of trucks and busses (f_{HV})

Vehicle type	Level of Service	Situation of the Land		
		Flat	Undulating	Mountainous
Truck (E_T)	A	2,0	4,0	7,0
	B and C	2,2	5,0	10,0
	D and E	2,0	5,0	12,0
Bus (E_B)	A	1,8	3,0	5,7
	B and C	2,0	3,4	6,0
	D and E	1,6	2,9	6,5

The service volume on multi-lane highways

The highways that have more than a lane for a direction and are built as divided or undivided are called multi-lane highways. The service volume can be calculated as:

$$Q_{SLi} = C_j \cdot (Q/C)_i \cdot N \cdot f_w \cdot f_{HV} \cdot f_e \cdot f_p$$

Q_{SLi} : The service volume of a single direction for the level of service (i) under dominant road and traffic conditions (veh/h)

C_j : The capacity per lane under ideal conditions for project speed (j). If the project speed ≥ 100 km/h, C_j=2.000 automobile/h/lane; if the project speed = 80 km/h, C_j=1.900 automobile/h/lane

$(Q/C)_i$: The volume/capacity rate for the level of service (i).

N : Number of lanes in a direction.

f_w : The adjustment factor about the lane and banquet width.

f_{HV} : The adjustment factor for heavy vehicle traffic.

f_e : The adjustment factor about the environmental conditions of the highway route.

f_p : The adjustment factor about the knowledge of the highway for the driver.

Level of service criteia on multi lane highways

Level of Service	Density (auto/km/ln)	Vp=110 km/h			Vp=100 km/h			Vp=80 km/h		
		$V_{avr.}$ (km/h)	Q/C	q_m (vh/h/ln)	$V_{avr.}$ (km/h)	Q/C	q_m (vh/h/ln)	$V_{avr.}$ (km/h)	Q/C	q_m (vh/h/ln)
A	≤ 7,5	≥ 91	0,36	700	≥ 80	0,33	650	-	-	-
B	≤ 12,5	≥ 85	0,54	1.100	≥ 77	0,50	1.000	≥ 67	0,45	850
C	≤ 18,8	≥ 80	0,71	1.400	≥ 70	0,65	1.300	≥ 62	0,60	1.150
D	≤ 26,3	≥ 64	0,87	1.750	≥ 64	0,80	1.600	≥ 56	0,76	1.450
E	≤ 41,9	≥ 48	1,00	2.000	≥ 48	1,00	2.000	≥ 45	1,00	1.900
F	> 41,9	< 48	v	v	< 48	v	v	< 45	v	v

v: Varies

Service volume on multi-lane highways

Level of Service	Unadjusted service volume (for a single lane (auto/h/ln)				
	Truck Percentage(%)				
	0	5	10	15	20
Flat Terrain					
A	700	700	650	650	650
B	1.100	1.050	1.000	1.000	1.000
C	1.400	1.350	1.300	1.250	1.250
D	1.750	1.700	1.650	1.600	1.550
E	2.000	1.950	1.850	1.800	1.750
Undulating Terrain					
A	700	600	550	500	500
B	1.100	950	850	800	700
C	1.400	1.200	1.100	1.000	900
D	1.750	1.500	1.350	1.250	1.100
E	2.000	1.750	1.550	1.450	1.250
Mountainous Terrain					
A	700	500	400	350	300
B	1.100	800	650	550	450
C	1.400	1.050	850	700	600
D	1.750	1.300	1.050	850	750
E	2.000	1.500	1.200	1.000	850

The adjustment factors about narrow lanes and banquets for multi-lane highways (f_w)

Distance to Obstacle (m)	Adjustment factors (f_w)							
	Obstacle on one side				Obstacle on both sides			
	Lane width (m)							
	3,7	3,3	3,0	2,7	3,7	3,3	3,0	2,7
Divided multi lane highway - Two lanes each side								
≥ 2,0	1,00	0,97	0,91	0,81	1,00	0,97	0,91	0,81
1,0	0,98	0,95	0,89	0,80	0,97	0,94	0,88	0,78
0	0,90	0,87	0,82	0,73	0,81	0,79	0,74	0,66
Divided multi lane highway - Three lanes each side								
≥ 2,0	1,00	0,96	0,89	0,78	1,00	0,96	0,89	0,78
1,0	0,98	0,94	0,88	0,77	0,97	0,93	0,86	0,76
0	0,94	0,91	0,85	0,74	0,91	0,87	0,81	0,70
Undivided multi lane highway - Two lanes each side								
≥ 2,0	1,00	0,95	0,89	0,77	N	N	N	N
1,0	0,97	0,93	0,87	0,76	N	N	N	N
0	0,88	0,85	0,80	0,70	0,81	0,79	0,74	0,66
Undivided multi lane highway - Three lanes each side								
≥ 2,0	1,00	0,95	0,89	0,77	N	N	N	N
1,0	0,98	0,94	0,88	0,76	N	N	N	N
0	0,94	0,90	0,83	0,72	0,91	0,87	0,81	0,70

* If the obstacle on each side is at different destination, the average is taken

N: The factors for the obstacle on one side is taken

Unit vehicle equivalents of the trucks and busses

Vehicle type	Terrain type		
	Flat	Undulaced	Mountainous
Truck (E_T)	1,7	4,0	8,0
Bus (E_B)	1,5	3,0	5,0

The adjustment factors represent the environmental properties (f_e)

Settlement type	Highway type	
	Divided	Undivided
Rural	1,00	0,95
Urban	0,90	0,80

Driver factors (f_p)

Driver type	(f_p)
Regular user	1,00
Occassional user	0,75-0,90

A new highway is designed by taking the level of service at the end of the service life into account. Parameters like the highway class (hence the project traffic and project speed), the land use properties of the route, and the topographical situation of the terrain are effective in the selection. The level of services for urban and suburban highways is given in the table.

These levels of services are to be considered as advice and should not be evaluated as design criteria.

The advised level of services at the end of service lives for suburban highways

Highway types	Terrain type		
	Flat	Undulaced	Mountainous
Multi lane highways	B	B	B - C
Two lane highways			
1st class highways	B	B - C	C
2nd class highways	C	C - D	D
3rd class highways	C	D	D
4th class highways	D	D	D

The advised level of services at the end of the service lives of urban highways

Highway types	Terrain type		
	Flat	Undulaced	Mountainous
Peripheral roads			
Multi lane highways	C	C	C
Two lane highways	C	C	C
Urban roads			
Multi lane highways	C	C	C
Two lane highways	C - D	C - D	C -D

5. CLASSIFICATION AND THE GEOMETRICAL STANDARDS OF HIGHWAYS

The highways have been classified according to some criteria, and the geometrical properties of each highway class have been defined. Classification has also brought standardization in time. Classification and geometrical standards aid engineers with their decisions on design. Classification and standardization are needed when considering this point of view. Similar standards have been used worldwide, while some countries use their own standards to represent their own conditions.

5.1 Classification of highways

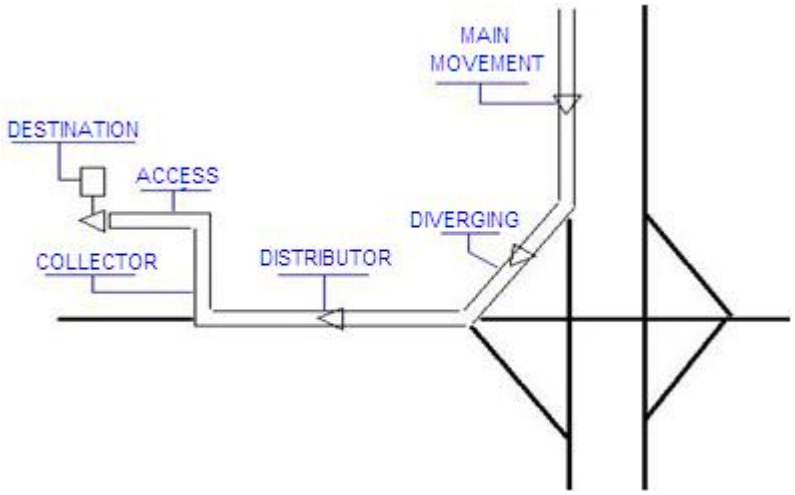
Highways can be classified according to some criteria, as seen below. Some highways can be listed in more than one class by definition.

- According to the properties of the zone that the highway passes from:
    - Urban highways: Highways surrounded by buildings, lighted, and have pedestrian traffic. They are divided into subcategories like arteries (speedways), boulevards, avenues, streets, and alleys.
    - Suburban highways: Suburban highway classes that are used in various countries are as follows:
        - 1st, 2nd, 3rd class highways.
        - Highway, expressway, speedway.
        - State highway, provincial road, rural road.
- According to the traffic value and geometry, it is the main road and the side road.
- According to the vehicle class that uses the highway, it includes automobile roads, pedestrian roads, bicycle roads, and bus roads.
- According to the type of traffic, it is residence road, travel road, and work-commerce road.
- According to the situation of the highway platform, it is divided road and undivided road.
- According to the flow direction of the traffic: One-sided road, two sided road.
- According to the type of pavement: Asphalt road, concrete road, stabilized road, unpaved road, parquet road.

The hierarchy of the movement

A totally functional design system generates a range of various transportation movements. 6 basic steps can be listed for many trips. These are main movement, transition, distribution, collection, access, and termination.

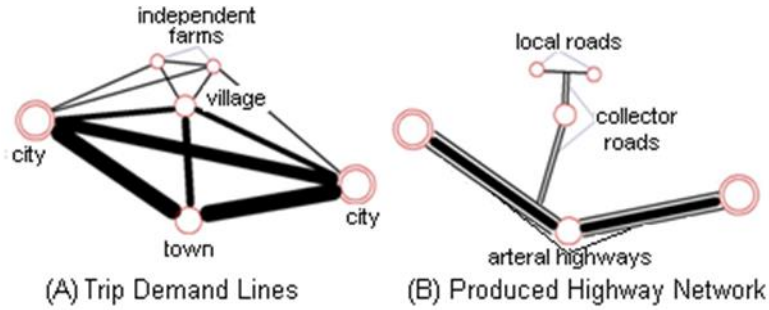
In the figure, a highway trip on which the vehicle movements are continuous and traffic flow is fast is shown. As vehicles diverge from the expressway, vehicles decrease their velocities. Then, vehicles join on main arterials, and a lower speed has to be selected.This is followed by collector roads. Then, vehicles reach local streets, which takes them to the last point of their trip. As they reach their last point, or destination, they park their cars in an appropriate car park. All steps of a typical trip fulfill its function by specially designed parts. As the hierarchy of the movement is related to the traffic load, expressway trips are usually on the highest level of the hierarchy. Distributer arterial, collector roads, and local roads will follow.



Functional relations

The function of a highway has to be defined to comprise the appropriate design criteria for classification of a highway. The design criteria aim to form a positive and consistent relation between the trip demands and the road network of residential areas, in other words, to optimize the system. Functional classification classifies the highways and the local ways according to their type of service. This classification shows that even the specific roads (streets and singular roads) do not serve independently. In other words, many roads are classified with respect to their trip purposes they serve. In this regard, the functional classification of roads is consistent with the classification of the trips.

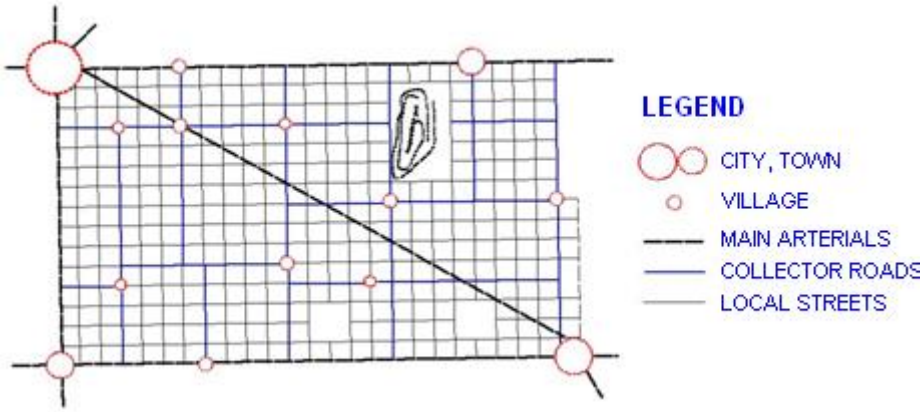
This main idea is shown schematically in the figure below. In Figure A, straight lines that connect the start and end points to each other symbolize the trip demand. The width of the lines symbolizes the size of the trip demand. The size of the circles symbolize the trip generation and attraction potential of the zones. As it's not easy to generate direct connections for each demand, trips are canalized on a limited road pattern, as shown in Figure B. The dense traffic movements are served directly, while lower-density movements are guided to indirect roads. The local streets, collector roads, and arterials on the figure are the descriptions that show the functional relations.



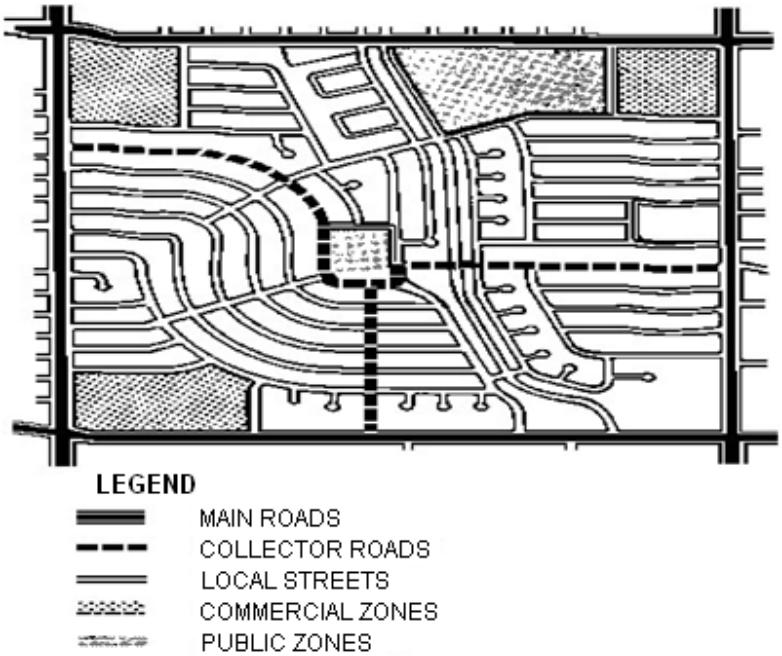
(A) Trip Demand Lines (B) Produced Highway Network

In the figure below, a more detailed schema of a functionally classified rural road network is shown. Main arterials usually provide a direct connection between cities and bigger towns while they generate longer trips. Mid-level collector roads directly serve the small cities and

connect them to the main arterial network. Local roads connect the independent houses or farms to the other local roads or to the collector roads.



The same approach is also valid for urban areas. A similar hierarchical system can also be defined for urban zones. However, the land use and the excess density of the trips complicate the determination of specific trip generation centers. In urban zones and suburbs, additional definitions like the distance of intersections have been more important for determining the appropriate and efficient network. In the figure below, a functionally classified suburban street network is shown schematically.

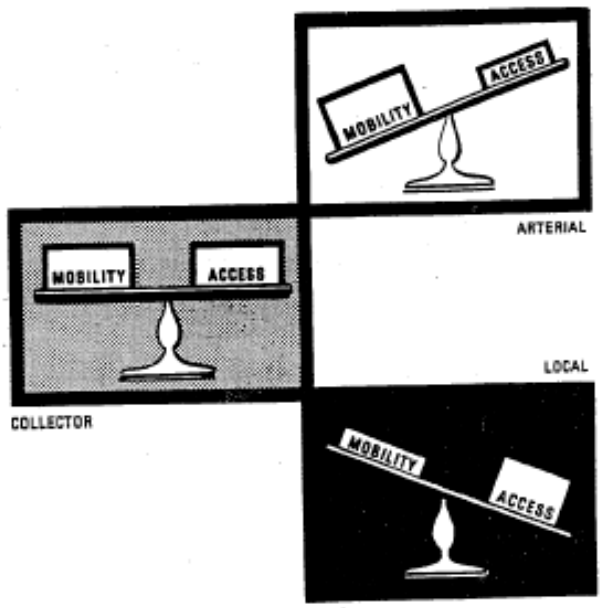


Access needs and controls

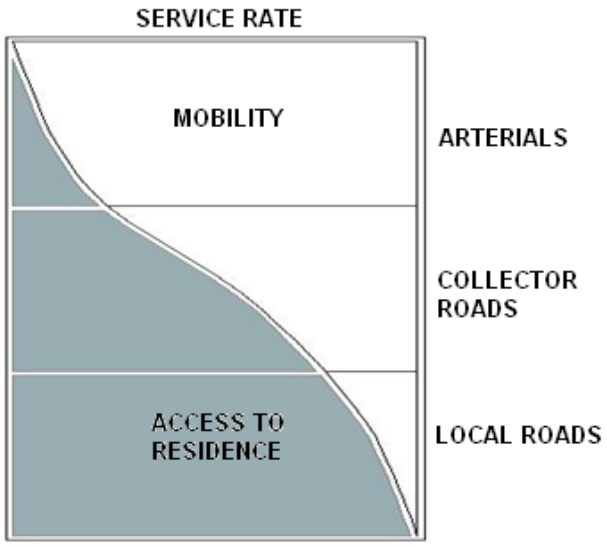
Two main factors in the classification of highway and street networks are accepted as access and mobility.

To preserve the continuity of the mobility and to provide access to various trip centers and targets coincides with each other.

The change of functions is shown by an equilibrium state shown in the figure below.



To provide the continuity of mobility, access restrictions have to be systematic on arterials. On the contrary, the main function of the local roads and streets is to provide access. The important factor in determining the functional class of streets or highways is the scope and degree of the access control. Access is a main need for every zone that is served by the road system. Mobility is composed of various levels of service and can be defined as drive comfort and lack of speed changes. The main factors are operating speed and the trip time.



The hierarchy of the trip distances is related to the access to house or work and the satisfaction of mobility needs, logically. Local rural settlements are the factors that determine the importance of access. The arterials show the importance of high mobility for the main movement and distribution. Collector roads provide balanced service for each function. In the figure above, this functional hierarchy is shown precisely.

As a result, the general descriptions below can be made with respect to the functional classification:

Arterials: They connect the major population centers nationally or internationally. On these roads, generation, and distribution traffic of major centers, as well as the traffic of collector roads, are present. They are high-standard roads. Divided roads, undivided roads, expressways, and highways are some examples of arterials.

Collector roads: They carry the generated and distributed traffic on smaller population centers than the arterials. The traffic and speed values are lower, so they have fewer standards. Country roads can be given as an example of collector roads.

Local roads: They usually connect to collector roads, grant access to the smaller population centers, and are the roads on which private estates and facilities are also connected. As being roads with the lowest standards, traffic, speed values, and trip comfort are low. The majority of a country's road network is composed of local roads.

Distribution of road types

An example from the USA. for the distribution of road types with respect to the functional classification is given below. Here, the maximum values of the ranges show the cities having less functional road networks. Lower values are seen in cities having more extensive road networks with respect to the population density.

System	RATIO	
	Trip Volume (%)	Length (%)
Main arterial system	40-65	5-10
Main arterial + secondary arterial	65-80	15-20
Collector road	5-11	5-10
Local road system	10-30	65-80

Systems	Ratio of the Total Rural Road Network
Main arterial system	2% - 4%
Main arterial + secondary arterial	6% - 12%, on most states 7% - 10%
Collector road	20% - 25%
Local road system	65% - 75%

5.2 Classification of Turkish Highways

The Turkish highways have been divided into four categories with respect to an administrative classification. They can be listed as:

1. Motorways
2. Stateways

3. Provincial roads
4. Country roads

The motorways, as the main roads, stateways, and provincial roads, as the collector roads, are the responsibility of the General Directorate of Highways. Country roads, as local roads, are under the responsibility of special provincial administrations; forest roads are under the responsibility of the Ministry of Environment and Forestry; urban roads are under the responsibility of local governments (municipalities).

Stateways can be in the form of divided or undivided roads with respect to their importance. Stateways can be subcategorized as first, second, or third-class roads.

Expressways, built to serve high traffic volume with high velocities, are constructed having at least 2 lanes in each direction, a total of 4, and divided with a central banquette. To provide a specific level of service, full access control is applied. This way, at grade intersections, pedestrians, bicycles, and animals do not exist on an expressway. To provide access control, obstacles are placed on each side of the road. Some expressways where intersections and entrances are allowed can be named partial access controlled roads.

Some of the main roads and collector roads have been enumerated with some codes in our country. Expressways are represented with the letter (O), while stateways are classified with the letter (D). Some of our highways are part of the international highway network. This kind of road is represented by the letter E.

Various roads with different codes can intersect with others at some sections.

- **Expressways:**
  - **Kapıkule-Ankara Expressway:**
    - (O-1) : Bosphorus Bridge pass and peripheral roads.
    - (O-2) : Fatih Sultan Mehmet Bridge pass and peripheral roads.
    - (O-3) : Kapıkule-İstanbul section,
    - (O-4) : İstanbul Ankara section,
    - (O-20) : Ankara peripheral road
  - **Aegean region expressways :**
    - (O-30) : İzmir peripheral road
    - (O-31) : İzmir-Aydın section
    - (O-32) : İzmir-Çeşme section
  - **Southern expressways:**
    - (O-21) : Adana-Pozantı section
    - (O-51) : Adana-Mersin section
    - (O-52) : Adana-Gaziantep section
    - (O-53) : Ceyhan-İskenderun section

- **Some stateways:**
  - **West-East direction:**
    - (D-100) : Kapıkule-İstanbul-Amasya-Erzurum-Gürbulak (Iran border)
    - (D-200) : Çanakkale-Bursa-Ankara-Sivas
    - (D-300) : Çeşme- Konya-Kayseri-Elazığ-Kapıköy (Iran border)
    - (D-400) : Datça-Antalya-Gaziantep-Esendere (Iran border)
  - **North-South direction:**
    - (D-550) : Uzunköprü-Çanakkale-İzmir-Aydın-Muğla
    - (D-650) : Karasu-Sakarya-Bilecik-Kütahya-Afyon-Burdur-Antalya
    - (D-750) : Zonguldak-Ankara-Aksaray-Mersin
    - (D-850) : Ünye-Tokat- Sivas-Adıyaman-Gaziantep-Kilis-Öncüpınar (Syria border)
    - (D-950) : Artvin-Erzurum-Bingöl-Diyarbakır-Mardin
- **European International Highway Network:**
- (E-80) : Kapıkule-İstanbul-Amasya-Erzurum-Gürbulak (Previously E-5)
- (E-87) : Aziziye (Kırklareli)-Çanakkale-İzmir-Antalya
- (E-90) : Çanakkale-Bursa-Ankara-Adana-Habur (Iraq border)
- **Important:** There isn't such highway that is named as E-6 in our country.

5.3 Geometrical standards

The examples of main geometrical standards of a highway are platform width (width of lanes and banquette), minimum horizontal and vertical curve radii, maximum grades and slopes, roof slope, bridge width, minimum underpass height, and right of way.

The selection of the geometrical standards is a matter of needs and costs.

For example, if a high volume of traffic needs to be passed fast and comfortably, geometrical standards have to be high. Also, highway standards have to provide the specific level of service stated in the project's life. The standards selected parallel to the selected level of service lead to the construction of an economical highway. In other words, the economy of a highway is not only related to the size but also the standards required by the selected level of service.

Geometrical standards that can be recommended for some road types

Road type	Terrain type	Project Vel. (km/h)	Min. curve radius (m)	Maksimum Grade (%)	Maks. grade Length (m)	Platform width (m)
Main road	Flat	80-110	190-360	4	No restriction	10-13
	Undulating	55-80	90-190	5-7	600 for >4%	10-13
	Mountainous	40-55	50-90	7-9	400 for >6%	8-10
Collector road	Flat	60-80	110-190	5	No restriction	10-12
	Undulating	50-60	75-110	5-7	No restriction	10-12
	Mountainous	35-50	35-75	7-9	750 for >9%	8-9
Local road	Flat	50-60	75-110	7	No restriction	7,5-8
	Undulating	35-50	35-75	7-9	No restriction	7,5-8
	Mountainous	25-35	30-35	9-12	1000 for >9%	7,5-8

Some parameters affecting the selection of geometrical standards can be listed as:

The class of the road: To serve a high traffic volume fast and comfortably, high standards are selected for main roads. The standards for collectors and local roads are lower.

Project speed: During the design of a highway, the maximum speed value that is essential to geometric design and the drivers can provide securely is named the project speed. This value has to be defined at the beginning of the design phase. Minimum sight distances, minimum horizontal curve radius, superelevation on the curves, transition curve length, widening distance, and the value of the comfort criterion are related to the project speed. As the project speed changes, these criteria also change, so it has an important effect. Project speed itself is not a geometrical standard. However, it is a design standard that affects the geometrical standards.

While determining the project speed, **(X)%** velocities are used. The recommended value is the **85% speed**. This speed value is the speed that 85% of the vehicles that pass from the road do not exceed. In other words, such a speed should be selected that at most 15% of the vehicles should exceed.

Also, the project speed may not be preserved through the route due to the topography and costs. In this case, the project speed on the relevant section is decreased, and a constraint speed is found. Constraint speed is used as the project speed in this section. However, a speed decrease to provide a balanced and secure trip should not be sudden but gradual. Recommended differences are at most 10~15 km/h between consequent sections.

Project traffic: The traffic values are the main values on the selection of the geometrical standards, as stated before. For this scope, the type of road and traffic amount that will pass from it during its service life is parallel to each other.

The vehicle composition on the traffic flow: Especially the heavy vehicle amount on the traffic flow affects the determination of the platform properties. On a highway having more heavy vehicles, lanes, and banquets are wider.

Topography of the terrain: The flat, undulating, or mountainous property of the terrain affects the road cost. On flat terrain, higher horizontal curve radii can be used, while on mountainous terrain, curve radii decrease. Similarly, on mountainous terrain, grades may not be decreased below a specific value. Even if the grade is decreased, high cut and fill costs can be encountered. For this reason, the type of terrain also has to be taken into account while project standards are being selected.

The economy: Determination of the geometrical standards is a result of the availability of financial possibilities. In addition to the needs, financial possibilities also affect the construction of a high-standard road. If the construction possibilities are limited, the road can be constructed gradually with respect to the project life and the level of service determined. For example, a road designed as (2x2) lanes can be constructed having 2 lanes in the beginning. According to the traffic increament amount, as enough right of way is determined, a lane can be added to both sides later on.

Safety: The main aim of the design is to pass a specific amount of traffic under a specific time and under specific conditions, economically, comfortably, and safely. Road and drive safety affects the selection of geometrical standards in order to prevent accidents. It is harder to provide safe driving conditions on low-standard roads that are narrow, have a high grade, limited sight distance, and have bad-conditioned pavement.

Other factors: While selecting the geometrical standards, the soil type, tectonic structure, climate, land use types, and natural and historical texture of the area also have to be considered.

Project components		Highway and (2x2) lane connection			
Project velocity		km/h	120	100	80
Lane width		m	3,75	3,75	3,75
Number of lanes			2x3	2x3	2x3
Banquet width (emergency parking lane)		m	3	3	3
Guardrail share (on fill)		m	1	1	1
Refuge banquet width		m	1	1	1
Refuge width		m	5	5	5
Normal slope (min. Superelevation)		%	2,5	2,5	2,5
Maximum superelevation		%	6	6	6
Minimum horizontal curve radius		m	1000	600	350
Minimum klotoid parameter		m	350	200	150
Horizontal curve radius (min. Superelevation)		m	3400	2050	1200
Horizontal curve radius that does not require a superelevation		m	5000	4500	3000
Horizontal curve radius that does not require a transition curve		m	3000	2000	1500
Allignment Length	Maximum	m	3000	2500	2000
	Minimum	m	600	500	400
Grade	Maximum	%	4	4	5
	Minimum	%	0,5	0,5	0,5
Relative slope	Maximum	%	0,5	0,5	0,5
	Minimum	%	0,3	0,25	0,25
Vertical curve (minimum) k = t/100 i	crest	m	200	125	70
	sag	m	100	70	35
Stop sight distance (desired),		m	300	200	150
Stop sight distance (minimum)		m	200	150	110
Minimum vertical span of a bridge		m	5	5	5

Stateway standards used by KGM - suburban

KARAYOLLARI GEOMETRİK STANDARTLARI (ÖZET TABLO - I)

TOPOĞRAFİK YAPI	TASARIM ELEMANLARI			KARAYOLU GEOMETRİK SINIFLARI									
				KENT DIŞI YOLLAR									
				ÇOK ŞERİTLİ YOLLAR (2x2)		İKİ ŞERİTLİ YOLLAR							
						1.SINIF		2.SINIF		3. SINIF		4.SINIF	
DÜZ	Tasarım Hızı		km/saat	100	90	100	80	80	70	70	60	50	40
	Hizmet Seviyesi	A,B,C,D,E,F		B	B	B	B	C	C	C	C	D	D
	Min DGM		m	180	150	180	125	125	100	100	80	65	50
	MinGGM		m	670	615	670	540	540	485	485	410	345	270
	Max Dever		%	8	8	8	8	8	8	8	8	8	8
	Enine Eğim		%	2.5	2.5	2	2	2	2	2	2	2	2
	Max Boyuna Eğim		%	4	4	4	4	5	5	6	6	10	10
	Min Kurp Yarıçapı		m	395	305	395	230	230	175	175	125	80	50
	Min Düşey Kurp	Açık	m	4400	3500	4400	2800	2800	2200	2200	1600	1300	900
	Yarıçapı	Kapalı	m	7400	5100	7400	3500	3500	2300	2300	1500	1000	600
	f_{max}			0.12	0.13	0.12	0.14	0.14	0.14	0.14	0.15	0.16	0.17
	Şerit Genişliği		m	3.5	3.5	3.5	3.5	3.5	3.5	3	3	3	3
	Banket Genişliği	İç	m	1	1								
		Dış	m	2.5	2.5	2.5	2.5	1.5	1.5	1	1	0.5	0.5
	K Katsayısı	Açık		44	35	44	28	28	22	22	16	13	9
	(DGM için)	Kapalı		73	51	73	35	35	23	23	14	10	6
	Orta Refüj Genişliği		m	4	4								
	Platform Genişliği		m	26*	26*	12	12	10	10	8	8	7	7
	Köprüler **		ton	Köprülerde gabari platform genişliği kadar olacaktır.									
	Kamulaştırma		m	Kamulaştırma genişliği projenin gerektirdiği kadar olcaktır.									
DALGALI	Tasarım Hızı		km/saat	90	80	80	70	70	60	60	50	40	30
	Hizmet Seviyesi	A,B,C,D,E,F		B	B	B-C	B-C	C-D	C-D	D	D	D	D
	Min DGM		m	150	125	125	100	100	80	80	65	50	30
	MinGGM		m	615	540	540	485	485	410	410	345	270	200
	Max Dever		%	8	8	8	8	8	8	8	8	8	8
	Enine Eğim		%	2.5	2.5	2	2	2	2	2	2	2	2
	Max Boyuna Eğim		%	5	5	6	6	7	7	8	8	12	12
	Min Kurp Yarıçapı		m	305	230	230	175	175	125	125	80	50	30
	Min Düşey Kurp	Açık	m	3500	2800	2800	2200	2200	1600	1600	1300	900	900
	Yarıçapı	Kapalı	m	5100	3500	3500	2300	2300	1500	1500	1000	600	600
	f_{max}			0.13	0.14	0.14	0.14	0.14	0.15	0.15	0.16	0.17	0.17
	Şerit Genişliği		m	3.5	3.5	3.5	3.5	3.25	3.25	3	3	2.75	2.75
	Banket Genişliği	İç	m	1	1								
		Dış	m	2.5	2.5	2.5	2.5	1.5	1.5	1	1	0.5	0.5
	K Katsayısı	Açık		35	28	28	22	22	16	16	13	9	4
	(DGM için)	Kapalı		51	35	35	23	23	14	14	10	6	2
	Orta Refüj Genişiliği		m	4	4								
	Platform Genişliği		m	26*	26*	12	12	9.5	9.5	8	8	6.5	6.5
	Köprüler **		ton	Köprülerde gabari platform genişliği kadar olacaktır.									
	Kamulaştırma		m	Kamulaştırma genişliği projenin gerektirdiği kadar olcaktır.									
DAĞLIK	Tasarım Hızı		km/saat	80	60	70	60	60	40	50	30	30	20
	Hizmet Seviyesi	A,B,C,D,E,F		B-C	B-C	C	C	D	D	D	D	D	D
	Min DGM		m	125	80	100	80	80	50	65	30	30	20
	MinGGM		m	540	410	485	410	410	270	345	200	200	200
	Max Dever		%	8	8	8	8	8	8	8	8	8	8
	Enine Eğim		%	2.5	2.5	2	2	2	2	2	2	2	2
	Max Boyuna Eğim		%	6	6	7	7	8	8	9	9	16	16
	Min Kurp Yarıçapı		m	230	125	175	125	125	50	80	30	30	10
	Min Düşey Kurp	Açık	m	2800	1600	2200	1600	1600	900	1300	900	900	900
	Yarıçapı	Kapalı	m	3500	1500	2300	1500	1500	600	1000	600	600	600
	f_{max}			0.14	0.15	0.14	0.15	0.15	0.17	0.16	0.17	0.17	0.18
	Şerit Genişliği		m	3.5	3.5	3.5	3.5	3.25	3.25	3	3	2.5	2.5
	Banket Genişliği	İç	m	1	1								
		Dış	m	2.5	2.5	2.5	2.5	1.5	1.5	1	1	0.5	0.5
	K Katsayısı	Açık		28	16	22	16	16	9	13	4	4	3
	(DGM için)	Kapalı		35	14	23	14	14	6	10	2	2	1
	Orta Refüj Genişliği		m	2	2								
	Platform Genişliği		m	23	23	12	12	9.5	9.5	8	8	6	6
	Köprüler **		ton	Köprülerde gabari platform genişliği kadar olacaktır.									
	Kamulaştırma		m	Kamulaştırma genişliği projenin gerektirdiği kadar olacaktır.									

*Çok şeritli yollarda otokorkuluk ve drenaj izgara payı gözönüne alınarak PG sağda ve solda 0.5m geniş tutulmuştur.

** Köprü hesaplarında kullanılan hareketli yükler "Yol Köprüleri için Teknik şartname" den alınacaktır.

Stateway standards used by KGM - urban

KARAYOLLARI GEOMETRİK STANDARTLARI (ÖZET TABLO - II)

TOPOĞRAFİK YAPI	TASARIM ELEMANLARI			KARAYOLU GEOMETRİK SINIFLARI							
				KENTSEL YOLLAR (KENT GEÇİŞLERİ)							
				ÇEVRE YOLLARI				KENT İÇİNDEN GEÇEN YOLLAR			
				ÇOK ŞERİTLİ		İKİ ŞERİTLİ		ÇOK ŞERİTLİ		İKİ ŞERİTLİ	
DÜZ	Tasarım Hızı		km/saat	100	60	90	60	80	60	70	50
	Hizmet Seviyesi	A,B,C,D,E,F		C	C	C	C	C	C	C-D	C-D
	Min DGM		m	180	80	150	80	125	80	100	65
	MinGGM		m	670	410	615	410	540	410	485	345
	Max Dever		%	8	8	8	8	4	4	4	4
	Enine Eğim		%	2.5	2.5	2	2	2.5	2.5	2	2
	Max Boyuna Eğim		%	4	4	4	4	4	4	4	4
	Min Kurp Yarıçapı		m	395	125	305	125	280	150	215	100
	Min Düşey Kurp	Açık	m	4400	1600	3500	1600	2800	1600	2200	1300
	Yarıçapı	Kapalı	m	7400	1500	5100	1500	3500	1500	2300	1000
	f_{max}			0.12	0.15	0.13	0.15	0.14	0.15	0.14	0.16
	Şerit Genişliği		m	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5
	Banket Genişliği	İç	m	1	1			1	1		
		Dış	m	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
	K Katsayısı	Açık		44	16	35	16	28	16	22	13
	(DGM için)	Kapalı		73	14	51	14	35	14	23	10
	Orta Refüj Genişiliği		m	4	4			4	4		
	Platform Genişliği		m	26*	26*	12	12	26*	26*	12	12
	Köprüler **		ton	Köprülerde gabari platform genişiliği kadar olacaktır.							
	Kamulaştırma		m	Kamulaştırma genişiliği projenin gerektirdiği kadar olcaktır.							
DALGALI	Tasarım Hızı		km/saat	80	60	80	60	70	50	60	30
	Hizmet Seviyesi	A,B,C,D,E,F		C	C	C	C	C	C	C-D	C-D
	Min DGM		m	125	80	125	80	100	65	80	30
	MinGGM		m	540	410	540	410	485	345	410	200
	Max Dever		%	8	8	8	8	4	4	4	4
	Enine Eğim		%	2.5	2.5	2	2	2.5	2.5	2	2
	Max Boyuna Eğim		%	5	5	6	6	5	5	6	6
	Min Kurp Yarıçapı		m	230	125	230	125	215	100	150	35
	Min Düşey Kurp	Açık	m	2800	1600	2800	1600	2200	1300	1600	900
	Yarıçapı	Kapalı	m	3500	1500	3500	1500	2300	1000	1500	600
	f_{max}			0.14	0.15	0.14	0.15	0.14	0.16	0.15	0.17
	Şerit Genişliği		m	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5
	Banket Genişliği	İç	m	1	1			1	1		
		Dış	m	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
	K Katsayısı	Açık		28	16	28	16	22	13	16	4
	(DGM için)	Kapalı		35	14	35	14	23	10	14	2
	Orta Refüj Genişiliği		m	4	4			4	4		
	Platform Genişliği		m	26*	26*	12	12	26*	26*	12	12
	Köprüler **		ton	Köprülerde gabari platform genişiliği kadar olacaktır.							
	Kamulaştırma		m	Kamulaştırma genişiliği projenin gerektirdiği kadar olcaktır.							
DAĞLIK	Tasarım Hızı		km/saat	80	60	60	50	60	40	60	30
	Hizmet Seviyesi	A,B,C,D,E,F		C	C	C	C	C	C	C-D	C-D
	Min DGM		m	125	80	80	65	80	50	80	30
	MinGGM		m	540	410	410	345	410	270	410	200
	Max Dever		%	8	8	8	8	4	4	4	4
	Enine Eğim		%	2.5	2.5	2	2	2.5	2.5	2	2
	Max Boyuna Eğim		%	6	6	7	7	6	6	7	7
	Min Kurp Yarıçapı		m	230	125	125	80	150	60	150	35
	Min Düşey Kurp	Açık	m	2800	1600	1600	1300	1600	900	1600	900
	Yarıçapı	Kapalı	m	3500	1500	1500	1000	1500	600	1500	600
	f_{max}			0.14	0.15	0.15	0.16	0.15	0.17	0.15	0.17
	Şerit Genişliği		m	3.5	3.5	3.5	3.5	3.5	3.5	3.5	3.5
	Banket Genişliği	İç	m	1	1			1	1		
		Dış	m	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
	K Katsayısı	Açık		28	16	16	13	16	9	16	4
	(DGM için)	Kapalı		35	14	14	10	14	6	14	2
	Orta Refüj Genişiliği		m	2	2			2	2		
	Platform Genişliği		m	23	23	12	12	23	23	12	12
	Köprüler **		ton	Köprülerde gabari platform genişiliği kadar olacaktır.							
	Kamulaştırma		m	Kamulaştırma genişiliği projenin gerektirdiği kadar olacaktır.							

*Çok şeritli yollarda otokorkuluk ve drenaj izgara payı gözönüne alınarak PG sağda ve solda 0.5 m. geniş tutulmuştur.

**Köprü hesaplarında kullanılan hareketli yükler "Yol Köprüleri için Teknik Şartname" den alınacaktır.

6. Highway route location

The route is the direction that a highway follows on the land. There are many possibilities to connect two points through a highway. The selection of the most appropriate route is named as the route study. Some subjects have to be paid attention during the route study. These are the design, construction, maintenance, repair, and pre-solution of potential problems that can be faced during the operation.

6.1 Route Location

6.1.1 Subjects that should be taken into account during the route study

- The route has to connect the main control points to each other. These are various.
- The route has to be appropriate for the standards that the selected highway class requires.
- The route has to serve the traffic expected to pass from the highway during its project life.
- The route has to be suitable for the main purpose of the highway. It shall satisfy the desired social, economic, industrial, commercial, and touristic purposes.
- The route has to pass from geologically suitable places.
- Route has to be selected in a way to decrease the earthwork costs.
- The route has to pass from places without drainage problems.
- The route has to be perpendicular to river passes in order to decrease the bridge costs.
- The route has to pass from places where the material suppliance is easy.
- Routes have to pass through places with low right-of-way costs.
- Routes have to pass through places with low maintenance costs (north-south difference).

It is hard for the route to fulfill all the requirements listed above. These conditions may conflict with each other. It is important to find the most appropriate selection.

6.1.2 Phases of the route study

To determine the most appropriate route for a highway whose main control points, standards, and traffic characteristics are specified, a study consisting of a preliminary survey, survey, and economic comparison has to be made.

6.1.2.1 Preliminary survey

During the preliminary survey phase, the terrain that the highway is expected to pass is examined, and potential feasible selections are determined. For this reason, topographical maps with a scale of 1/25.000 and geological maps with a scale of 1/100.000 are used. The feasible selections determined on the map are examined topographically, geologically, and geotechnically on the terrain. The first eliminations are made accordingly. First, the selections that;

- Increase the cost in terms of earthworks and engineering structures,
- Geologically infeasible,
- High construction and maintenance costs, topographically

eliminated. The sensibility of the preliminary study changes with respect to the highway class. A report for each remaining selection is prepared. On the report,

- The topographical situation of the terrain,
- The geological formation, geotechnical structure, and landslide condition through the route,
- Condition of the underwater and surface waters and also the drainage possibilities,
- Situation of the potential material quarries,
- The place, type, approximate sizes, and costs of potential engineering structures,
- Route length
- Right of way conditions
- A rough quantity survey and a general cost analysis are present.

The first thing to do while determining the highway route is the **zero line study**. This study followed between necessary points on a topographical map, acts as a guide for the route to be passed appropriate to the **selected standards**.

Zero line study

If the elevations of the two necessary points, A (start) and C (end), are known and the distance between them can be approximated (with the air distance or rope polygon method), a highway can be constructed between these points with a specific grade. The grade acquired with this method is:

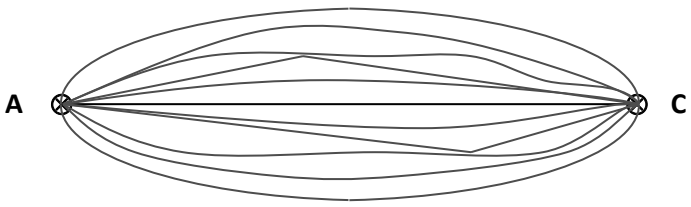
$$g_0 = \frac{\Delta h_{A-C}}{L_{A-C}} \cdot 100$$

g_o : The grade of the highway (%)

$\Delta h_{A\text{-}C}$: The elevation difference between the necessary points (m)

$L_{A\text{-}C}$: The approximate distance between the necessary points (m)

However, the question of where the highway route shall pass can not be answered precisely. Many alternatives arise with respect to the two parameters affecting this decision: the **distance between the points** and the **grade**. Also, alternatives are infinite, depending on the terrain type, which may be flat, undulating, or mountainous. Basically, the zero-line study is made to reduce the number of alternatives to a more reasonable number.



The **zero line** is placed between the consequent contour lines. By connecting the zero lines, a broken line (polygon) is obtained. The earthwork obtained under this condition is called the zero polygons, and the work done is named the **zero line study**.

If a profile is drawn using the elevations of the points where the zero polygon intersects with the contour lines, a straight line with a constant grade is obtained. The grade of the zero line has to be lower than the maximum grade (g_{max}) given on the project standards.

The grade of a zero polygon is the same as the zero lines. In other words, if the grade of the total is constant, pieces with the same length should also be constant and the same. Accordingly, the grade between the consequent contour lines can be defined as:

$$\text{tg}\,\alpha = g_0 = \frac{h}{l}\cdot 100$$

h : Elevation difference between the consequent contour lines (m)

l : Length of the zero line (m)

α : The angle of the zero line with the horizontal plane (angle)

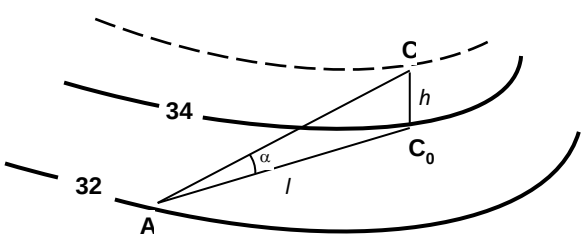
The length of the zero line can be written as:

$$l = \frac{h}{g_0}\cdot 100 = \frac{h}{\text{tg}\,\alpha}$$

The elevation difference between the two consequent contour lines changes with respect to the scale of the map. On 1/1.000 scale 1 m, 1/2.000 scale 2 m and 1/5.000 scale 5 m is used. For lower-scale maps (like 1/10.000, 1/25.000), different values can be used. It is recommended to start with an approximate grade for the zero-line study.

To determine the measure of the starting grade with respect to the maximum grade, the distance between the necessary points can be calculated using the air distance or rope polygon). To apply the zero line length calculated on the map, it should be converted into the scale of the map. The length, converted into the scale of the map, is the distance between the legs of the compasses. The distance between the legs of the compasses (l_c) for a specific scale:

$$l_c = l\cdot(scale)$$



Example:

Find the distance between the legs of the compasses for the zero line study followed on a map with a scale of 1/2.000 to connect the necessary A and C points which have an elevation difference of 40 m and a distance of 900 m between them.

Solution: $g_0 = \frac{\Delta h_{A-C}}{L_{A-C}} \cdot 100 = \frac{40}{900} \cdot 100 = 4{,}44\%$ $l = \frac{h}{g_0} \cdot 100 = \frac{2}{4{,}44}\ 100 = 45\text{m}$

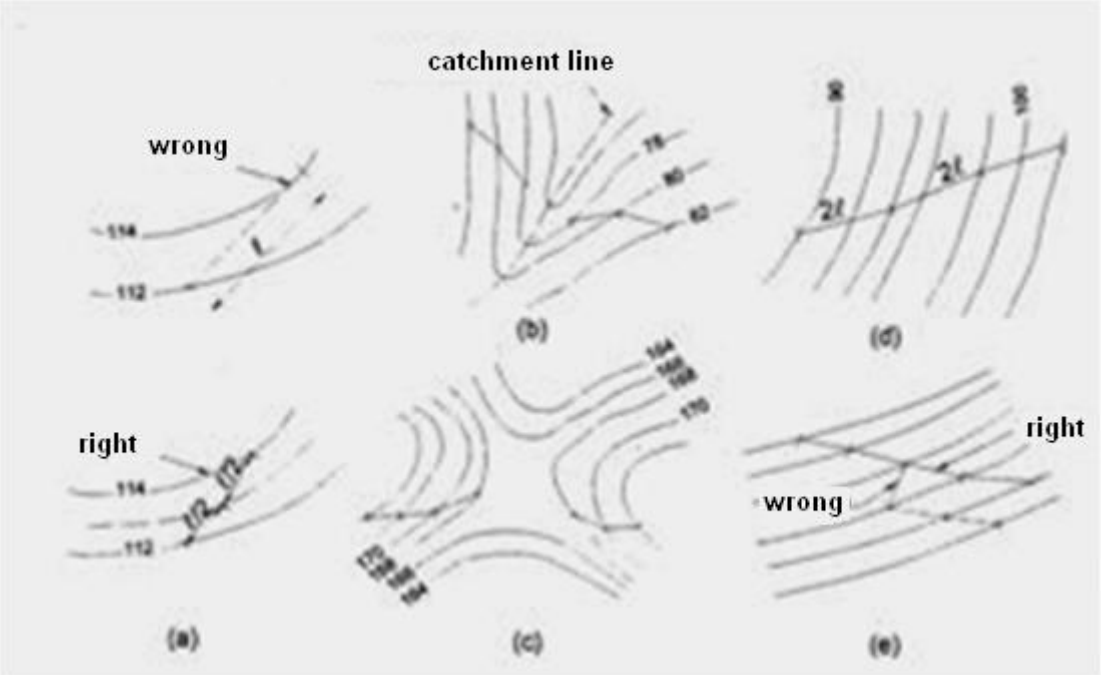
$l_c = l \cdot (scale) = 45\ (\frac{1}{2.000}) = 0{,}0225\text{m} = 2{,}25\text{cm}$

After the length of the zero line is determined, one leg of the compass is placed on the contour line at the start of the highway, and the next contour line, with respect to the processing direction, is intersected. The intersection point is connected with the start point using a ruler. This operation will continue until the endpoint is reached. The polygon obtained at the end of the consequent operations is named the **zero polygon**.

The following matters have to be taken into account during the zero-line study:

- A single grade should be used as much as possible between the necessary points. If needed, the grade of the zero polygon can be changed, and various grades can be used in different sections. However, in this condition, declines and inclines should be avoided until being forced to.
- If the second necessary point can not be connected using the calculated distance between the legs of the compasses, the grade selected is not appropriate. For this reason, the grade should be changed, and the new distance between the legs of the compasses should be calculated. To increase the distance between the legs of the compasses, the grade should be decreased or vice versa.
- The zero line must intersect with the contour lines only once and clearly. If it becomes tangent on ridges and valleys, the distance between the consequent contour lines should be divided into two (into four if needed), and a gradual pass must be made. During this process, imaginary contour lines are placed between the two consequent contour lines. The distance between the legs of the compasses should be decreased with the same ratio of the divisions. For example, the length of the zero line is halved on two graded passes (Figure a).
- On valley base floors and passes of the hills, contour lines having the same elevation can be skipped, and the polygon can be drawn broken (Figure b-c).
- On urgent cases, one of the two contour lines can be skipped to gain time. In this case, the zero line length should be taken as 2*l* (Figure d). The skipped contour lines can be increased as needed.
- On flat terrain, sharp turns should be avoided if possible.

Various conditions encountered during the zero-line study



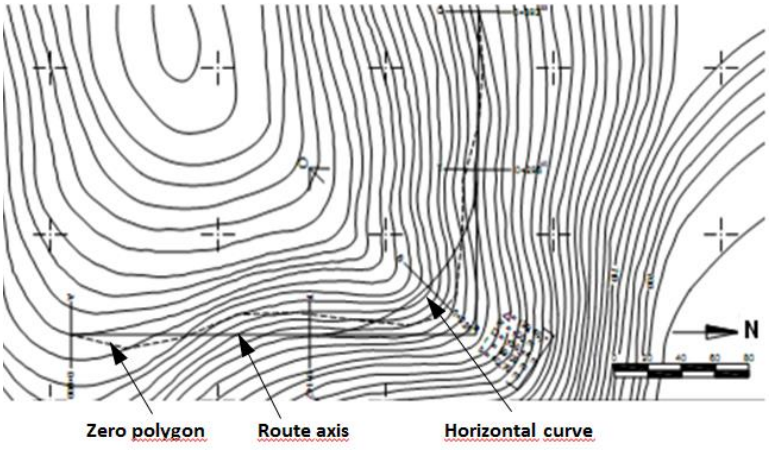
Determining the road axis

The vehicles can not move safely and comfortably under desired project standards on a route composed of various broken lines. For this reason, the zero polygon, composed of various broken lines, has to be transformed into a less broken route axis. As the route axis is placed between the necessary points, it will inevitably diverge from the zero polygon. The highway type and project standards determine the degree of the divergences. A high divergence is not permitted on a low-standard highway. The route is placed as close as possible to the zero polygon while adapting to the project standards in order to decrease the amount of earthworks.

On high-standard highways, divergences from the zero polygons can be accepted to obtain good travel conditions. The important point is that the amount of earthwork is going to increase as divergence from the zero polygon increases. In the zero polygon study, as the divergences from the zero polygon are inevitable, the earthwork should be balanced, or in other words, cuts and fills should be balanced on the economical haul distances. The road axis is composed of **line segments** and **curves**. The line segments are named as the **alignments**. The circle arcs placed between the alignments are named the **curves**. The other curves used

for various purposes are named the **transition (connection) curves**. These are special curves in the form of a **klotoid, cubical parabola, lemniscat,** and the **circle arc with a radius of 2R.**

In the figure below, a route example placed on the zero polygon is shown.



6.1.2.2 Survey

More detailed research on the feasible routes selected after the preliminary survey phase is followed during the survey phase. In this phase, topographical surveys and soil surveys are made in general.

A map with a 1/25.000 scale is not precise enough to determine the route. Topographical maps with higher scales have to be produced about the route alternatives remaining after the preliminary survey. According to the importance of the highway, contour maps for suburban highways with a scale of 1/5000 or 1/2.000 are produced for route lines of 100-300 m. For expressways, the scale is selected as 1/1.000, and the route lines are around 700-800 m.

The routes drawn during the preliminary survey phase are transferred to the new maps drawn after the necessary corrections are made. The plan, the profile, and the cross-sections are prepared for each route alternative. After this phase, a preliminary project is obtained. On the maps prepared, engineering structures, intersections with the other highways, material suppliance places, highway facilities, and private properties have to be shown.

Soil surveys occupy an important role in the survey phase. Detailed geological and geotechnical surveys are prepared for each route alternative. Boreholes are opened in specific places and within specific distances during these surveys. The samples obtained are sent to laboratories for research. The underground water level also has to be researched.

A comparison, and if needed, another elimination, is followed after the data is gathered in the survey phase. This way, the alternatives for the economic comparison are confirmed.

6.1.2.3 Economical comparison

The last comparison made to determine a certain route is the economic comparison. Using the data gathered, information about the various highway cost items is obtained. It is possible to find the most efficient route alternative with an economical comparison in light of these factors. Various economic analysis methods have been developed for this subject, such as the **net present value, internal rate of return,** and the **benefit/cost ratio**. As highway investments also have a social side, usually, the **benefit/cost ratio** method is selected.

For the methods listed above, equivalent amounts on highway investments and the other costs and benefits of the highway are calculated. Therefore, the monetary values of each income and expenditure item have to be calculated as the monetary value for a specific year in order to avoid the effects of the changes in economic conditions. According to the economic rules, in a specific inflation condition, the present value of the money is always higher than the future value. This change in the value of the money can be resolved by the actualization process. Therefore, the investment costs are actualized to a future date when the highway is put into service with a specific interest rate, while the operation and maintenance-repair costs are also actualized to a previous date when the highway is put into service with a discount rate. The discount process expresses the inverse of the interest process. For actualization, the compound interest formula is used:

$$A_n = A_0 \quad (1+i)^n$$

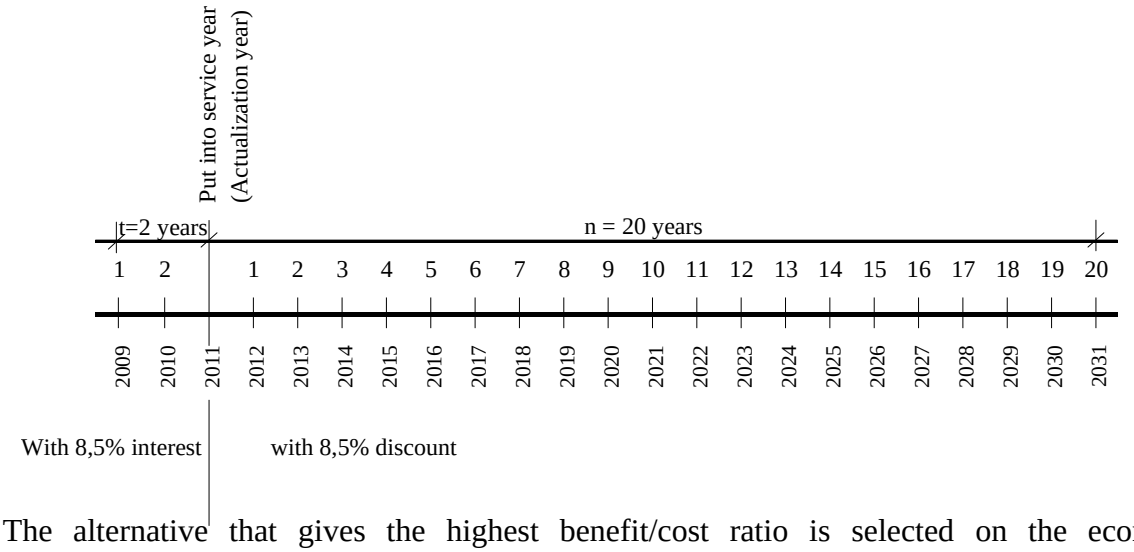
A_n : The equivalent of the capital A_0 with the interest rate of *i* on *n* years later.

A_0 : Present equivalent of a capital.

By using this formula, it is also possible to find the equivalent of a present capital on *n* years before with the interest rate of *i*. This process is named as the discount and *i* is named as the discount rate. Discounted value (A′) can be written as the opposite of the previous formula:

$$A' = \frac{A_0}{(1+i)^n}$$

Thus, the benefit/cost ratio can be calculated by dividing the actualized total benefits by the actualized total costs. As this ratio increases, the efficiency, benefit, or, in other words, the profit of the investment increases. Transportation investments have a social aspect due to their nature. The profit is not only stated in cash. The social, cultural, and industrial development of a region, the investment of the highway into the region, and the employment possibilities should also be considered as long-term profits. When viewed from this aspect, many direct and indirect benefit and cost items are present.



The alternative that gives the highest benefit/cost ratio is selected on the economic comparison of a highway. Under normal conditions, this ratio is desired to be higher than 1. The alternative having the highest benefit/cost ratio (between the ones higher than 1) is selected. However, on some conditions, for the development of a region or because of strategic reasons, investment programs whose benefit/cost ratio is below 1 can also be applied. In Turkey, highways are considered in investment programs of developing regions, this condition is encountered. The most common period of assessment is 20 years, based on the economic comparison.

As stated in the service life, the highways are expected to end their economic life at the end of this period after providing the desired conditions. The highway, which ends its economic life, continues its service. However, to improve the lower level of service on the last condition, the highway has to be improved, or additional facilities have to be built.

6.1.3 Application

After the economic comparison, the most suitable route is selected for construction.

- Points of intersections of horizontal curves are determined (if these points are determined on the map with a scale of 1/2000 during the study phase, it is called as the **study application**; else if they are determined directly on the land, it is called as the **direct application**).
- Picketing (Determining the alignments and the curves on the land).
- Levelling/Application levelling is done by holding levelling rods on the stakes direven on the land during the picketing phase. The profile is obtained after this phase.
- Determining cross-sections (Cross-sections are taken to specify the situation of the land vertical to the highway route).

6.2 Other studies about the route

- Preparation of the local dimensional drawings (1/5.000, 1/1.000)
- Preparation of the nationalization plan (1/2.000). If the land has not been registered in a cadastre, it is prepared by KGM employees otherwise is prepared by KGM and General Directorate of Land Registry and Cadastre employees together.
- Setting of slope stakes
- Preparation of the route plan